

BitVLA: 1-bit Vision-Language-Action Models for Robotics Manipulation

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Abstract

Vision-Language-Action (VLA) models have shown impressive capabilities across a wide range of robotics manipulation tasks. However, their growing model size poses significant challenges for deployment on resource-constrained robotic systems. While 1-bit pretraining has proven effective for enhancing the inference efficiency of large language models with minimal performance loss, its application to VLA models remains underexplored. In this work, we present **BitVLA**, the first 1-bit VLA model for robotics manipulation, in which every parameter is ternary, i.e., $\{-1, 0, 1\}$. To further reduce the memory footprint of the vision encoder, we propose the distillation-aware training strategy that compresses the full-precision encoder to 1.58-bit weights. During this process, a full-precision encoder serves as a teacher model to better align latent representations. Despite the lack of large-scale robotics pretraining, BitVLA achieves performance comparable to the state-of-the-art model OpenVLA-OFT with 4-bit post-training quantization on the LIBERO benchmark, while consuming only 29.8% of the memory. These results highlight BitVLA's promise for deployment on memory-constrained edge devices. We release the code and model weights in <https://github.com/ustcwhy/BitVLA>.

1 Introduction

Recent years have witnessed remarkable progress in vision-language models (VLMs) [HLG⁺24, ZWC⁺25, WBT⁺24, BCL⁺25]. These models have achieved impressive results across a wide range of downstream tasks, such as visual question answering [LLWL23, LLLL24], mathematical reasoning [ZHY⁺25, WQH⁺25], and human-agent interaction [HWL⁺24, QYF⁺25]. Building upon this progress, the field is increasingly moving toward vision-language-action (VLA) models, which extend the modalities of VLMs to incorporate action generation for robotic control [BBC⁺23, ZYX⁺23, DXS⁺23, MZH⁺23, KPK⁺24, LLZ⁺24]. These models aim to endow robots with the ability to understand visual environments, follow natural language instructions, and perform tasks autonomously. VLA models offer a unified framework to bridge perception, language understanding, and motor control, making them a promising paradigm for embodied AI.

However, deploying such large-scale VLA models in real-world robotic systems remains highly challenging, particularly on resource-constrained edge devices. These systems are often limited in terms of memory, computational throughput, and energy availability. Recent efforts in model quantization have shown that reducing the bit-width of model weights and activations can yield substantial improvements in efficiency. In particular, 1-bit large language models (LLMs) [WMD⁺23, MWM⁺24, MWH⁺25], where every parameter is restricted to ternary values (i.e., $\{-1, 0, 1\}$), have emerged as a compelling solution. These models achieve competitive performance on a variety of NLP benchmarks while dramatically reducing memory footprint, energy consumption, and inference latency. Moreover, the ternary parameter space enables efficient hardware execution and can simplify

deployment on edge accelerators. Despite their promise, existing 1-bit models have been largely confined to the language domain. To the best of our knowledge, their extension to multimodal tasks and robotic control has not yet been thoroughly explored.

In this work, we introduce **BitVLA**, the first 1-bit vision-language-action model for robotics manipulation, where every parameter is ternary, i.e., $\{-1, 0, 1\}$. BitVLA is built upon the publicly available 1-bit LLM BitNet b1.58 2B4T [MWH⁺25]. We begin by training a vision-language model using the 1-bit LLM in conjunction with a full-precision vision encoder, following the training paradigm of LLaVA [LLWL23]. To further reduce memory footprint, we introduce distillation-aware training to quantize the vision encoder to 1.58-bit weights and 8-bit activations. During this stage, we only train the vision encoder of the model, in which the full-precision encoder is used as the teacher model to better align the latent representations. Despite the absence of large-scale robotics pretraining, as shown in Figure 1, BitVLA achieves performance on par with the state-of-the-art model OpenVLA-OFT [KFL25] with 4-bit post-training quantization, while only using 29.8% memory footprint. These results demonstrate that BitVLA offers a cost-effective and high-performance solution for robotics manipulation, making it feasible for memory-constrained robotic systems.

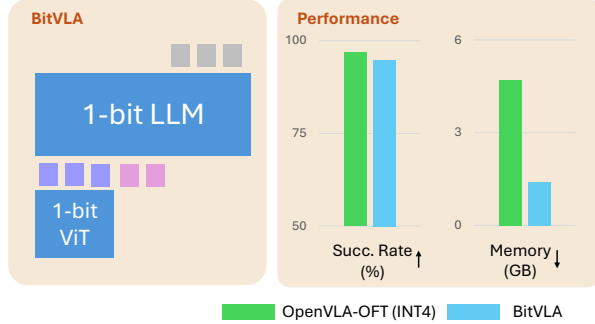


Figure 1: Comparison between BitVLA and OpenVLA-OFT with 4-bit post-training quantization in terms of end task performance and memory footprint. We report the average success rate on LIBERO benchmark.

2 Related Works

Vision-Language-Action models. Inspired by the rapid progress of VLMs, researchers in robotics have begun exploring VLA models that directly generate low-level control signals. The RT series [ZYX⁺23, ORM⁺] introduced Open X-Embodiment (OXE), a large-scale standardized robotics dataset, and used it to train RT-X, a generalist model for robotic manipulation tasks. OpenVLA [KPK⁺24] provided a detailed discussion on the design of VLA, covering aspects from the pretraining architecture to parameter-efficient fine-tuning methods and deployment strategies, while fully open-sourcing the training methods across all stages and the pre-trained model. RoboFlamingo [LLZ⁺24] leveraged pre-trained VLMs for single-step vision-language reasoning, introduced a policy head to capture sequential history, and required minimal fine-tuning via imitation learning. OpenVLA-OFT [KFL25] optimized the fine-tuning process by modeling continuous actions, employing parallel decoding, and applying action chunking from imitation learning [ZKLF23, CFD⁺23]. To improve inference efficiency, TinyVLA [WZL⁺24] adopts a compact 1.3B VLM backbone and skips pretraining to enhance data efficiency. Most recently, NORA [HSH⁺25] demonstrated competitive performance by utilizing Qwen2.5-VL-3B [BCL⁺25] as its backbone, enhanced with the FAST+ tokenizer for action generation.

Native 1-bit models. Modern deep learning research is increasingly focused on quantization-aware training and low-precision inference [PWW⁺23, XLCZ23, LLH⁺23]. Recent studies [WMD⁺23, MWM⁺24, KVM⁺24, ZZS⁺24, WMW24, WMW25] have demonstrated the potential of 1-bit and 1.58-bit pre-training for LLMs. [WMD⁺23] empirically showed that the performance gap between 1-bit and full-precision models narrows as the parameter count increases. Further, BitNet b1.58 [MWM⁺24] showed that 1.58-bit LLMs can match the performance of full-precision models starting from the 3B scale, while significantly reducing inference costs in terms of memory footprint, decoding latency, and energy consumption. OneBit [XHY⁺24] further explored the use of knowledge distillation for training binary LLMs. bitnet.cpp [WZS⁺25] developed an inference system optimized for 1-bit LLMs, substantially lowering energy consumption and improving decoding latency on CPU devices. More recently, [MWH⁺25] trained a 2B-parameter ternary LLM, achieving competitive performance relative to leading open-weight LLMs. The low memory and energy require-

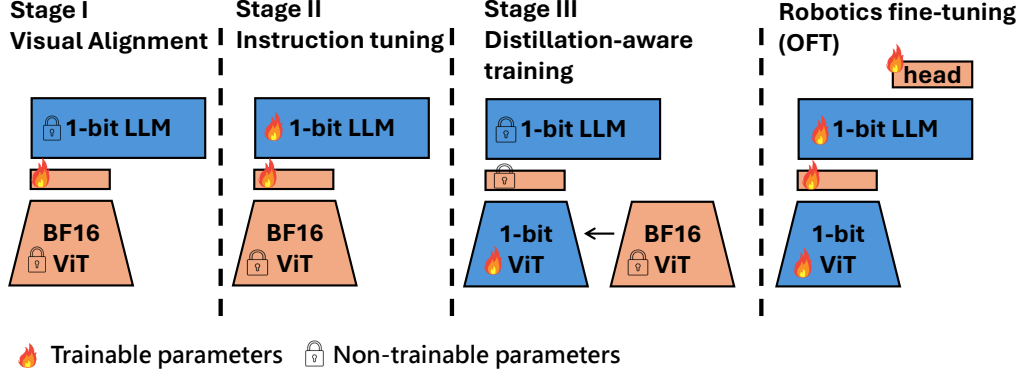


Figure 2: The overview of the training of BitVLA. We first train a vision-language model using a 1-bit LLM [MWH⁺25] combined with a full-precision vision encoder. Then we apply distillation-aware training to quantize the vision encoder’s weights to 1.58-bit precision. Finally, BitVLA is adapted to specific robotics tasks through OFT fine-tuning [KFL25].

ments of 1-bit LLMs make them particularly attractive for edge applications, especially for robotics tasks. However, to the best of our knowledge, the extension of 1-bit models to vision-language and vision-language-action training remains largely unexplored.

3 BitVLA: 1-bit VLA

In this section, we first introduce the model architecture of BitVLA in Section 3.1. Next, we detail the objectives of the distillation-aware training strategy in Section 3.2, which aims to compress the full-precision encoder weights to 1.58-bit. Finally, we describe the fine-tuning procedure of BitVLA on downstream tasks in Section 3.3.

3.1 Model Architecture

BitVLA adopts BitNet b1.58 2B4T [MWH⁺25] as the LLM backbone and utilizes SigLIP-L [ZMKB23] as the vision encoder. We employ the version of SigLIP-L pre-trained on 224×224 resolution images, resulting in shorter sequences of visual tokens. This choice enhances computational efficiency with minimal impact on performance [KPK⁺24]. We use a two-layer MLP with GeLU activation functions as the connector, which remains in full-precision due to its negligible contribution to the overall model size.

Figure 2 provides an overview of the training of BitVLA. Following the training strategy of LLaVA [LLWL23], we begin by training the VLM using the 1-bit LLM and a full-precision vision encoder. In the first stage, only the connector is trained on a small image captioning dataset to align the LLM with the vision encoder. This is followed by a visual instruction tuning phase, during which the vision encoder is frozen while the remaining components are updated. Finally, we perform distillation-aware tuning to quantize the vision encoder into a low-bit representation.

For quantization, we employ the *absmean* quantizer for weights and the per-token *absmax* quantizer for activations [MWM⁺24]. The weights are quantized to ternary values (i.e., $\{-1, 0, 1\}$), while the activations are quantized to symmetric INT8 (i.e., $[-128, 127]$). Specifically, the quantization can be formulated as:

$$Q_w(W) = \alpha \cdot \text{RoundClip}\left(\frac{W}{\alpha}, -1, 1\right), \alpha = \frac{1}{nm} \|W\|_1 \quad (1)$$

$$Q_a(x) = \frac{\beta}{127} \cdot \text{RoundClip}\left(\frac{127x}{\beta}, -128, 127\right), \beta = \|x\|_\infty \quad (2)$$

$$\text{RoundClip}(x, a, b) = \max(a, \min(b, \text{round}(x))) \quad (3)$$

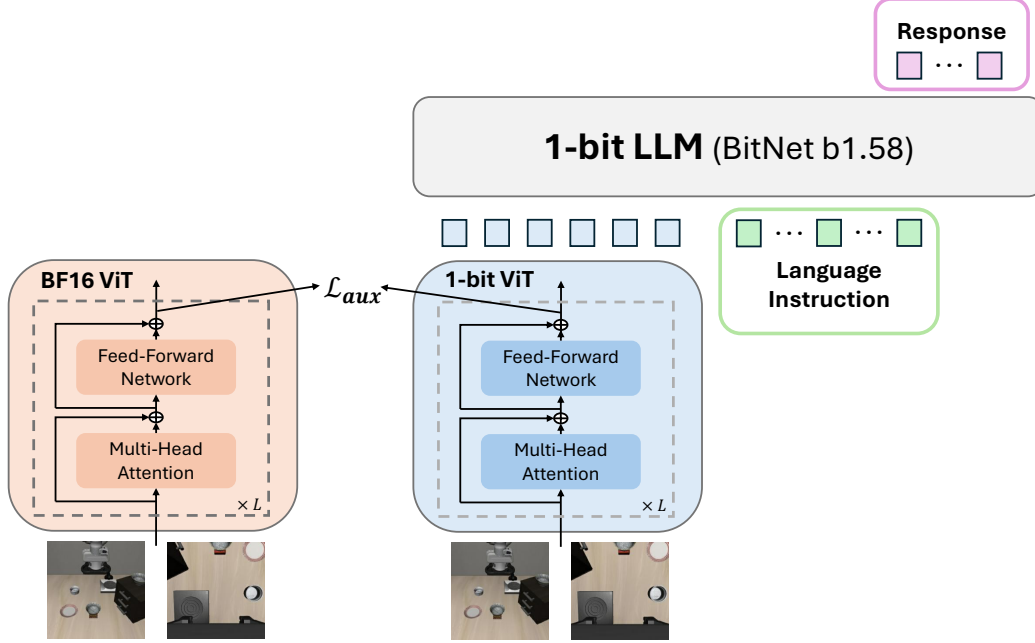


Figure 3: The overview of the distillation-aware training. The original full-precision encoder serves as the teacher model to ensure better alignment of the latent representations.

where $W \in \mathcal{R}^{m \times n}$ denotes the learnable weight of linear layer and $x \in \mathcal{R}^{n \times 1}$ denotes the inputs. The output of a ternary linear layer is computed as $Y = Q_w(W)Q_a(x)$, where Q_w and Q_a denote the quantization functions for weights and activations, respectively.

We apply quantization to all linear layers in the vision encoder, excluding the input and output embedding layers. BitVLA is trained with quantization-aware training, where quantization is performed on-the-fly during the forward pass. Due to the non-differentiable nature of quantization operations, we adopt the straight-through estimator (STE) [BLC13] to approximate gradients during backpropagation. Specifically, the gradients are passed directly through the quantization functions, following the approximation:

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial Q_w(W)}, \quad \frac{\partial \mathcal{L}}{\partial X} = \frac{\partial \mathcal{L}}{\partial Q_a(X)} \quad (4)$$

Both the gradients and optimizer states are maintained in full precision to preserve training stability.

3.2 Distillation-aware Training

In this subsection, we introduce the distillation-aware training to effectively quantize the vision encoder of VLM to 1.58 bit-widths. We illustrate the overview in Figure 3. We first initialize the latent weights of 1.58-bit encoder from its full-precision counterpart. Then we adopt the full-precision encoder as the teacher model. The training objective $\mathcal{L}_{total}$ requires the minimization of both task-specific loss $\mathcal{L}_{LM}$ and an auxiliary alignment loss $\mathcal{L}_{aux}$ of latent representations between full-precision and 1.58-bit encoder.

Language modeling loss. The auto-regressive language modeling loss, $\mathcal{L}_{LM}$, is widely used in training VLMs. Let $\mathcal{T}$ denote the input text sequence, which is divided into an instruction part $\mathcal{T}_{ins}$ and a response part $\mathcal{T}_{ans}$. The visual tokens extracted by the 1.58-bit vision encoder are denoted as $\mathcal{V}_{1.58\text{-bit}}$. The language modeling loss can be formulated as:

$$\mathcal{L}_{LM} = - \sum_{\text{token}_i \in \mathcal{T}_{ans}} \log \Pr(\mathcal{Y}^i | \mathcal{V}_{1.58\text{-bit}}, \mathcal{T}^{[:i-1]})$$

where $\mathcal{Y}^i$ represents the model’s predicted token at position i . The loss is computed only over the response tokens $\mathcal{T}_{\text{ans}}$, while the instruction and visual tokens are provided as context.

Representations alignment loss. To enhance the alignment between the latent representations of the 1.58-bit and full-precision vision encoders, we learn the 1.58-bit encoder through knowledge distillation, in which the full-precision encoder is used as the teacher model. Let h_{bf16}^l and $h_{1.58\text{-bit}}^l$ denote the outputs of the l -th layer from the full-precision and 1.58-bit vision encoders, respectively. The alignment loss is defined as:

$$\mathcal{L}_{\text{aux}} = \frac{1}{n} \sum_{l=1}^L \|h_{\text{bf16}}^l - h_{1.58\text{-bit}}^l\|^2$$

where n is the hidden dimension and L is the total number of layers in the vision encoder. This auxiliary loss encourages the 1.58-bit vision encoder to mimic the representational behavior of its full-precision counterpart.

Above all, the training objective $\mathcal{L}_{\text{total}}$ is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \gamma \cdot \mathcal{L}_{\text{aux}} \quad (5)$$

where γ is a coefficient for representations alignment. During the distillation-aware training, only the vision encoder is trainable while the other components (i.e., LLM and connector) are frozen. In our experiments, we observe that, unlike the 1.58-bit pre-training of LLMs, the quantization-aware training of 1.58-bit encoder is highly data-efficient with distillation from a full-precision teacher model. It preserves most of the performance of its full-precision counterpart using only billions of training tokens.

3.3 Robotics Fine-tuning

In this subsection, we describe the fine-tuning procedure of BitVLA for specific robotics tasks. Following OpenVLA-OFT [KFL25], we utilize parallel decoding and action chunking techniques to enhance the throughput of VLA models. Specifically, we replace the conventional causal mask used in LLMs with a bidirectional attention mask, enabling each forward pass to generate a coherent action trajectory over multiple time steps. This approach significantly boosts real-time control efficiency compared to autoregressive, token-by-token predictions. Additionally, we integrate an MLP-based action head to project the latent representations of query tokens into continuous robotic action space. The model is trained to minimize the L_1 loss between predicted actions and ground-truth trajectories.

4 Experiments

4.1 Model Training

BitVLA is trained with a three-stage procedure. Following LLaVA [LLWL23], we first train the connector to align the vision encoder with the LLM using the LLaVA 1.5-558k dataset [LLLL24]. In the second stage, we freeze the vision encoder and train both the LLM and the connector on a 10-million-sample subset of MammoTH-VL [GZB⁺24], consisting of single-image samples. In the final stage, we train the vision encoder from full-precision (W16A16) to 1.58-bit weights and 8-bit activations (W1.58A8) on a 5-million-sample subset from the data in the second stage. The training data in Stage III contains up to 10B tokens. The distillation loss on latent representations is weighted by a coefficient $\gamma = 0.1$. As recommended by [MWM⁺24], we use a large learning rate for instruction tuning. The training requires 14 days on 8 NVIDIA A100 cards with 80GB memory. We present the detailed hyperparameter configurations in Appendix A.

4.2 Experiments on Robotics Manipulation

Benchmark. We adopt the LIBERO simulation environment [LZG⁺23] to evaluate the generalization and performance of robotics manipulation models. As shown in Figure 4, this benchmark assesses robotic intelligence across four critical dimensions: spatial generalization (manipulating

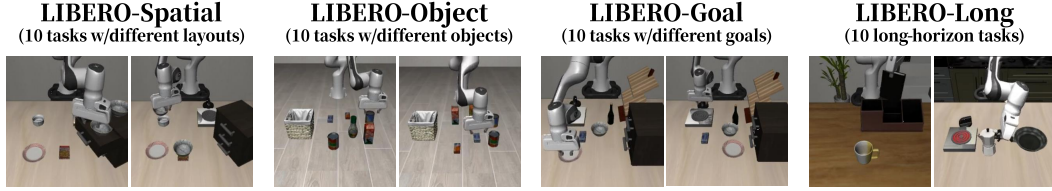


Figure 4: The overview of LIBERO benchmark task suites. It has four different dimensions to evaluate the generalization and performance of robotics manipulation models.

Table 1: The success rate (%) of BitVLA and the baselines on LIBERO simulation environment.

Models	Size	Memory Usage↓	Spatial	Object	Goal	Long	Avg.
<i>w/ Robotics pre-training</i>							
OpenVLA [KPK ⁺ 24]	7.5B	15.1GB (10.79×)	84.7	88.4	79.2	53.7	76.5
SpatialVLA [QSC ⁺ 25]	4.2B	8.5GB (6.07×)	88.2	89.9	78.6	55.5	78.1
CoT-VLA [ZLK ⁺ 25]	8.0B	16.2GB (11.57×)	87.5	91.6	87.6	69.0	81.1
NORA-Long [HSH ⁺ 25]	3.8B	7.5GB (5.36×)	92.2	95.4	89.4	74.6	87.9
π_0 [BBD ⁺ 24]	3.5B	7.0GB (5.00×)	96.8	98.8	95.8	85.2	94.2
OpenVLA-OFT [KFL25]	7.7B	15.4GB (11.00×)	97.6	98.4	97.9	94.5	97.1
<i>w/o Robotics pre-training</i>							
OpenVLA-OFT [KFL25]	7.7B	15.4GB (11.00×)	94.3	95.2	91.7	86.5	91.9
BitVLA (ours)	3.0B	1.4GB (1.00×)	97.4	99.6	94.4	87.6	94.8

objects arranged in novel configurations), object generalization (adapting to previously unseen object categories), goal generalization (interpreting diverse language instructions), and long-horizon reasoning (performing multi-stage tasks involving varied objects, layouts, and objectives). These capabilities are systematically evaluated through four corresponding task suites, namely LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long. Each task suite contains 500 expert demonstrations systematically distributed across 10 distinct manipulation tasks. Additional details are included in Appendix B.

Implementation details. We use the same training dataset¹ as OpenVLA-OFT [KFL25] during fine-tuning. We process synchronized multi-view visual inputs from both wrist-mounted and external cameras, while encoding proprioceptive signals such as end-effector positions. The physical state measurements are projected into a single token using an MLP-based projector, which is then appended to the image tokens. For action chunking, we set chunk size to $K = 8$ following OpenVLA-OFT, and execute full chunks before re-planning.

We perform full-parameter fine-tuning for faster convergence across all experiments. Specifically, BitVLA is fine-tuned for 10k steps on LIBERO-Spatial, LIBERO-Object, and LIBERO-Goal, and for 100k steps on LIBERO-Long. We adopt a cosine decay learning rate schedule with a batch size of 64. The 10k-step fine-tuning process takes approximately 4 hours on 8 NVIDIA A100 cards with 80GB of memory. More details can be found in Appendix A.

Baselines. We compare BitVLA with the baselines under supervised fine-tuning on the LIBERO dataset, including OpenVLA-OFT [KFL25], OpenVLA [KPK⁺24], SpatialVLA [QSC⁺25], CoT-VLA [ZLK⁺25], NORA-Long [HSH⁺25], and π_0 [BBD⁺24]. Specifically, π_0 employs the flow-matching architecture built upon a pre-trained VLM. NORA is trained from a strong lightweight VLM Qwen2.5-VL-3B [BCL⁺25] to improve the efficiency. We adopt its NORA-Long variant, which generates five-step action sequences at a time. CoT-VLA introduces visual chain-of-thought reasoning by predicting future frames autoregressively before action generation. SpatialVLA incorporates 3D information and learns a generalist policy for manipulation. OpenVLA is a 7B open-source VLA model trained on the OXE dataset, surpasses closed models like RT-2-X [ORM⁺] in numerous tasks. OpenVLA-OFT is fine-tuned from OpenVLA using a series of techniques, e.g., parallel decoding and continuous action modeling, to improve the speed and performance for specific task. Due to

¹https://huggingface.co/datasets/openvla/modified_libero_rlds

Table 2: The success rate (%) of BitVLA and OpenVLA, OpenVLA-OFT with post-training quantization on LIBERO simulation environment.

Models	Memory Usage↓	Spatial	Object	Goal	Long	Average
<i>INT8 post-training quantization</i>						
OpenVLA [KPK ⁺ 24]	7.4GB (5.29×)	86.4	85.2	77.2	58.8	76.9
OpenVLA-OFT [KFL25]	7.7GB (5.50×)	98.8	98.0	96.6	94.0	96.7
<i>INT4 post-training quantization</i>						
OpenVLA [KPK ⁺ 24]	4.4GB (3.14×)	83.0	84.0	72.0	51.6	72.7
OpenVLA-OFT [KFL25]	4.7GB (3.36×)	98.2	98.2	97.2	93.8	96.9
BitVLA (ours)	1.4GB (1.00×)	97.4	99.6	94.4	87.6	94.8

Table 3: The zero-shot accuracy of BitVLA with full-precision and 1.58-bit vision encoder (VE) on visual question answering tasks.

Models	MMMU (val)	SeedBench (image)	SeedBench ²⁺ (test)	MMStar (test)	AI2D (test)	Avg.
BitVLA w/ 16-bit VE	37.4	70.6	45.0	43.6	68.6	53.0
BitVLA w/ 1.58-bit VE	35.4	69.3	43.7	41.5	67.6	51.5

resource constraints, BitVLA is not pre-trained on large-scale robotics datasets. Therefore, we also report OpenVLA-OFT results fine-tuned directly from its base VLM [KNB⁺24] for reference.

Main results. Table 1 summarizes the success rates of BitVLA and various baselines on the LIBERO benchmark suites. As shown in the table, although BitVLA is not pre-trained on a large-scale robotics dataset (e.g., Open X-Embodiment [ORM⁺]), it still surpasses strong baselines with 3 billion parameters, including π_0 and NORA-Long. In particular, BitVLA outperforms π_0 by 2.4% on LIBERO-Long, highlighting its effectiveness in long-horizon reasoning tasks for robotic manipulation. Moreover, BitVLA has a lightweight memory footprint of only 1.4GB, making it feasible to deploy on a single consumer-grade GPU such as the NVIDIA GeForce RTX 3050 Ti Laptop (4GB).

Compared to the larger OpenVLA-OFT model, BitVLA achieves comparable performance on the Spatial, Object, and Goal subsets of the LIBERO benchmark, but still falls short on LIBERO-Long. We attribute this gap to OpenVLA-OFT’s fine-tuning from OpenVLA, which benefits from large-scale robotics pre-training and thus excels at complex manipulation tasks. As shown in the table, pre-training on a large-scale robotics dataset boosts OpenVLA-OFT’s success rate on LIBERO-Long from 86.5% to 94.5%. Notably, when compared to the OpenVLA-OFT variant without robotics pre-training, BitVLA achieves comparable performance on LIBERO-Long.

Comparison with post-training quantization. We compare BitVLA with OpenVLA and OpenVLA-OFT models subjected to 8-bit and 4-bit post-training quantization. We use their publicly available fine-tuned checkpoints released on Hugging Face. For quantization, we employ the bit-sandbytes toolkit [DLBZ22] to convert the model backbones to INT8 and INT4 precision. We report both the memory footprint and performance of the quantized models on the LIBERO benchmark. As shown in Table 2, OpenVLA exhibits a larger performance degradation under 4-bit quantization compared to OpenVLA-OFT. Notably, BitVLA achieves performance comparable to 4-bit quantized OpenVLA-OFT while using less than one-third of the memory.

4.3 Experiments on Visual Question Answering

We evaluate the zero-shot performance of BitVLA with both full-precision and 1.58-bit vision encoders on visual question answering (VQA) tasks. The evaluation suite includes MMMU [YNZ⁺24], SeedBench [LWW⁺23], SeedBench-2-Plus [LGC⁺24], MMStar [CLD⁺24], and AI2D [KSK⁺16]. We adopt the publicly available LMM-Eval toolkit [ZLZ⁺24] to ensure fair and consistent comparisons. As shown in Table 3, BitVLA equipped with the 1.58-bit encoder achieves performance comparable to its full-precision counterpart. Specifically, the 1.58-bit encoder results in only a 1.5%

Table 4: The ablations on data size and representation alignment loss of distillation-aware training for visual question answering tasks.

Training Tokens (Stage III)	$\mathcal{L}_{aux}$	MMMU (val)	SeedBench (image)	SeedBench ²⁺ (test)	MMStar (test)	AI2D (test)	Avg.
10B	✓	35.4	69.3	43.7	41.5	67.6	51.5
5B	✓	33.3	69.1	43.3	41.4	66.4	50.8
5B	✗	32.4	52.9	38.8	30.7	57.5	42.4

Table 5: The ablations on data size and representation alignment loss of distillation-aware training on LIBERO benchmark suites.

Training Tokens (Stage III)	$\mathcal{L}_{aux}$	Spatial	Object	Goal	Long	Average
10B	✓	97.4	99.6	94.4	87.6	94.8
5B	✓	96.8	98.6	93.8	85.2	93.6
5B	✗	96.2	98.6	91.4	85.2	92.9

average accuracy drop across the five benchmarks, while reducing its memory footprint from 0.8GB to 0.1GB. These results demonstrate that distillation-aware training effectively preserves performance on general VQA tasks while significantly lowering memory consumption during inference.

4.4 Ablation Studies

Representations alignment loss. We perform ablation studies to assess the impact of the proposed representation alignment loss during distillation-aware training. As shown in Table 4, incorporating the alignment loss significantly boosts the zero-shot performance of BitVLA with the 1.58-bit vision encoder, raising the average accuracy from 42.4% to 50.8% across five VQA benchmarks. On the LIBERO benchmark suites, where models are fine-tuned for specific tasks, the performance gain is smaller but still meaningful. As reported in Table 5, the alignment loss leads to a 2.4% increase on LIBERO-Goal set.

Data size of distillation-aware training. We compare the performance of BitVLA trained with 5B and 10B tokens during the distillation-aware training (Stage III). As shown in Table 4, increasing the training data during the quantization-aware training of the vision encoder improves overall performance on general VQA tasks. Specifically, BitVLA trained with 10B tokens in Stage III surpasses the 5B-token counterpart by 0.7% in average accuracy. Additionally, on the LIBERO benchmark, the 10B-token model achieves a 1.2% gain in average accuracy after fine-tuning.

5 Qualitative Analysis

In this section, we carefully analyze the failure cases of BitVLA on LIBERO benchmark suites, categorizing them into three types: spatial localization discrepancy, goal misunderstanding, and trajectory planning failure.

- **Spatial localization discrepancy** refers to the failure caused by inaccuracies in pose prediction during manipulation. Typically, such errors occur in four main scenarios: (1) manipulating objects with unstable centers of gravity, such as wine bottles, where small miscalculations lead to instability (2) selecting imprecise grasping poses, causing objects to topple during approach or drop during transportation (3) phantom manipulation attempts in the absence of physical objects (4) positioning errors when placing objects at target locations, resulting in task failures. These issues may arise from either coarse spatial understanding derived from visual encoder or over-reliance on proprioceptive signals relative to visual information under certain operational conditions in BitVLA.
- **Goal misunderstanding** refers to failure arising from BitVLA’s incorrect interpretation or following of the language instructions. A typical scenario of this error occurs when the robot

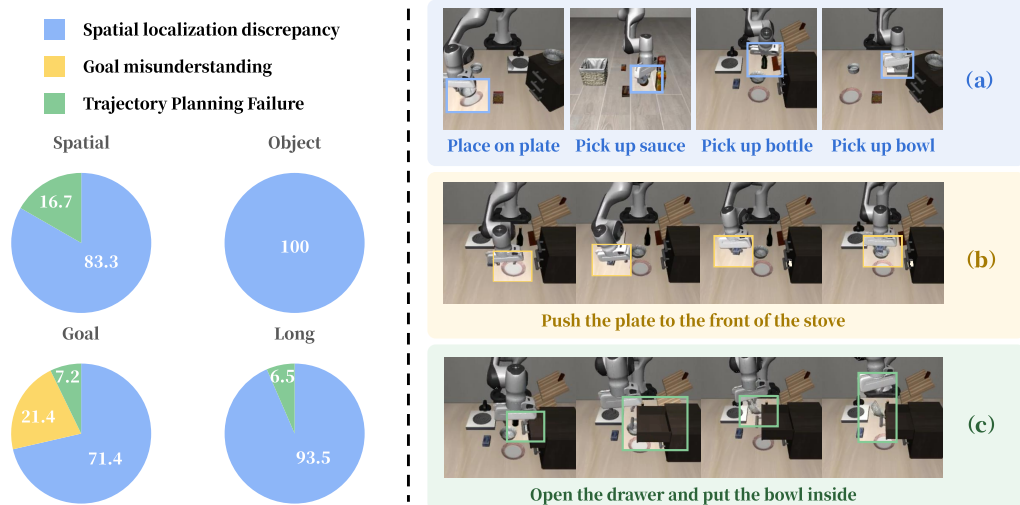


Figure 5: **Left:** distribution of failure types across each task suite on the LIBERO simulation environment. **Right:** typical examples for (a) spatial localization discrepancy, (b) goal misunderstanding and (c) trajectory planning failure.

erroneously interacts with non-target objects during task execution, subsequently initiating another task rollout associated with the contacted objects. We hypothesize that this primarily arises from the dominance of visual and proprioceptive information during the model’s reasoning process at the moment of goal switching. This misalignment is also mentioned in OpenVLA-OFT [KFL25], which proposes OpenVLA-OFT+ augmented variant employing a FiLM strategy to alleviate this issue.

- **Trajectory Planning Failure** refers to execution errors caused by collisions during motion planning. A typical case involves the robotic arm colliding with the lower panel of an open drawer during bowl placement, leading to bowl drops or the arm becoming jammed. These failures highlight the need for BitVLA to better leverage prior knowledge for generating more rational and collision-free trajectories. Moreover, the system should anticipate the feasibility of subsequent sub-goals (e.g., placing a bowl) when executing earlier ones (e.g., opening a drawer) to avoid operational conflicts. For instance, partially opening the drawer may reduce trajectory complexity and lower the risk of collision.

We illustrate the distribution and specific examples for each failure type in Figure 5. The most frequent failure type across all task suites is spatial localization discrepancy. We found that the success criteria of LIBERO are very strict, especially in placement tasks, where success is only determined if the object is placed precisely in the center of the plate. However, in many failure cases within the Long suite, objects were successfully placed on the plate but still failed because they were not centered. Nevertheless, the large number of errors in this category indicates that dexterous manipulation tasks remain the biggest bottleneck for BitVLA.

6 Conclusion

We present BitVLA, the first 1-bit vision-language-action model for robotics manipulation, where every parameter is constrained to ternary values. BitVLA is initially trained using a 1-bit LLM and a full-precision vision encoder. To further compress the vision encoder, we introduce a distillation-aware training strategy that converts its weights to ternary values. In this stage, the full-precision encoder serves as the teacher model to guide the alignment of latent representations. Experimental results on the LIBERO benchmark show that BitVLA achieves performance on par with the state-of-the-art OpenVLA-OFT model with 4-bit post-training quantization, while reducing memory usage by up to $3.36\times$. These results highlight BitVLA as a cost-effective and efficient solution for robotics applications on memory-constrained hardware.

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A Hyper-parameters

We present the hyperparameter configurations used for training BitVLA in Table 6. Following the recommendations of [MWM⁺24], we employ a two-stage weight decay schedule during visual instruction tuning. For fine-tuning on the LIBERO-Spatial, LIBERO-Object, and LIBERO-Goal suites, we report the best results selected from learning rates in the set 5e-5, 1e-4, 3e-4. For LIBERO-Long, all models are trained with a peak learning rate of 8e-5 for the vision encoder and 4e-4 for the LLM.

Table 6: Hyper-parameters for the training of BitVLA.

Hyper-parameter	Stage I	Stage II	Stage III
Peak Learning rate	1e-3	3e-4	1e-4
Batch Size	256	256	256
Weight decay	$\times$	0.1 $\rightarrow$ 0	0.01
Trainable modules	Connector	LLM, Connector	ViT
Training steps	25k	40k	20k
Training sequence	1024	2048	2048
Vision sequence		256	
Learning rate scheduling		polynomial decay	
AdamW β		(0.9, 0.999)	
AdamW ϵ		1e-8	
Gradient Clipping		1.0	
Dropout		$\times$	
Attention Dropout		$\times$	

Table 7: Hyper-parameters for the fine-tuning of BitVLA on LIBERO dataset.

Hyper-parameter	Spatial	Object	Goal	Long
Peak Learning rate	{5e-5, 1e-4, 3e-4}			4e-4, 8e-5
Training steps	10k	10k	10k	100k
Learning rate scheduling		cosine decay		
Warmup steps		375		
Batch Size		64		
Weight decay		0.01		
Trainable modules		LLM, Connector, ViT		
AdamW β		(0.9, 0.999)		
AdamW ϵ		1e-8		
Gradient Clipping		$\times$		

B Tasks in LIBERO

In this section, we present the detailed task compositions of each task suite in LIBERO. As shown in Table 8, it demonstrates the distinct task configurations across the four task suites within the LIBERO framework. Figure 6 illustrates the scene visualizations for a subset of tasks.

Table 8: Task description in LIBERO benchmark task suites.

Task suite	Task description
Spatial	pick up the black bowl between the plate and the ramekin and place it on the plate pick up the black bowl next to the ramekin and place it on the plate pick up the black bowl from table center and place it on the plate pick up the black bowl on the cookie box and place it on the plate pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate pick up the black bowl on the ramekin and place it on the plate pick up the black bowl next to the cookie box and place it on the plate pick up the black bowl on the stove and place it on the plate pick up the black bowl next to the plate and place it on the plate pick up the black bowl on the wooden cabinet and place it on the plate
Object	pick up the alphabet soup and place it in the basket pick up the cream cheese and place it in the basket pick up the salad dressing and place it in the basket pick up the bbq sauce and place it in the basket pick up the ketchup and place it in the basket pick up the tomato sauce and place it in the basket pick up the butter and place it in the basket pick up the milk and place it in the basket pick up the chocolate pudding and place it in the basket pick up the orange juice and place it in the basket
Goal	open the middle drawer of the cabinet put the bowl on the stove put the wine bottle on top of the cabinet open the top drawer and put the bowl inside put the bowl on top of the cabinet push the plate to the front of the stove put the cream cheese in the bowl turn on the stove put the bowl on the plate put the wine bottle on the rack
Long	put both the alphabet soup and the tomato sauce in the basket put both the cream cheese box and the butter in the basket turn on the stove and put the moka pot on it put the black bowl in the bottom drawer of the cabinet and close it put the white mug on the left plate and put the yellow and white mug on the right plate pick up the book and place it in the back compartment of the caddy put the white mug on the plate and put the chocolate pudding to the right of the plate put both the alphabet soup and the cream cheese box in the basket put both moka pots on the stove put the yellow and white mug in the microwave and close it

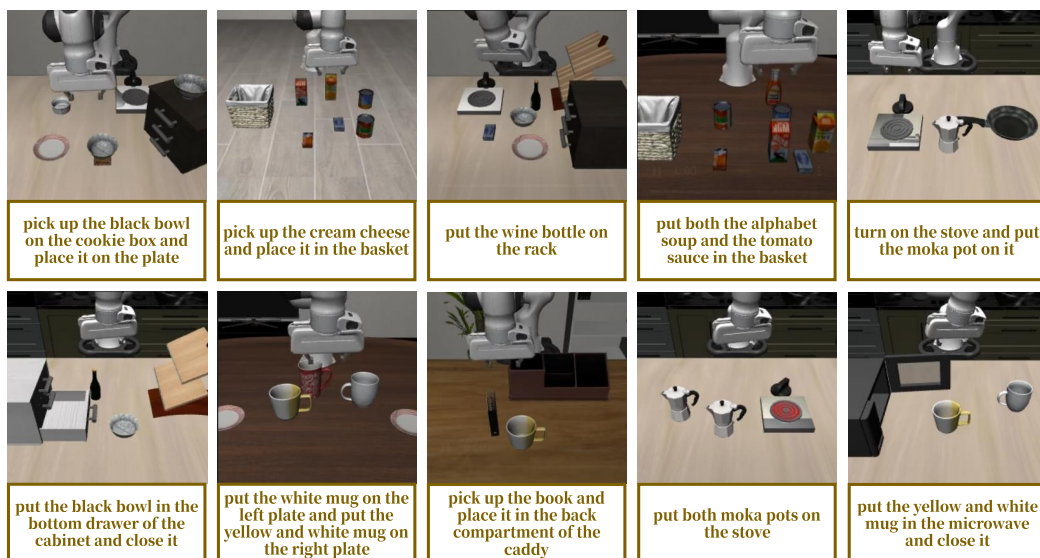


Figure 6: Examples in LIBERO benchmark tasks suites.